Smart irrigation system using IoT and machine learning methods

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***Abstract*—Water loss and improper scheduling are problems with traditional irrigation techniques, making it difficult to meet the growing demand for food production while also preserving precious water resources. To address these challenges, this paper introduces a cutting-edge smart irrigation system that leverages the power of the Internet of Things (IoT), data analysis, and machine learning to determine the most efficient way to apply and schedule water for irrigation. The cornerstone of this innovative system is the utilization of a carefully curated dataset, "Crop Irrigation Scheduling," sourced from Kaggle. This dataset comprises six crucial attributes: Crop Type, Crop Days, Soil Moisture, Temperature, Humidity, and Irrigation. These attributes, described in detail in the dataset's metadata, provide the foundational information required for precise irrigation management. The system operates seamlessly by deploying IoT sensors to collect real-time data from the field. This data is then preprocessed to ensure its quality and consistency. Subsequently, a machine learning model is trained using this dataset to make intelligent irrigation decisions in real time. The model's predictions are seamlessly integrated with control mechanisms that govern the irrigation process. Moreover, the system offers an intuitive API for easy data access and management, allowing agricultural professionals to monitor and adjust irrigation strategies effortlessly. Experimental findings validate the system's utility, with particular emphasis on the decision tree model, which demonstrates the best balance of accuracy, precision, recall, and F1 score. This system maximizes crop yield and minimizes water waste, contributing to a more sustainable and productive agricultural ecosystem.**

Keywords—Smart irrigation system, machine learning methods, smart agriculture, IoT.

I. Introduction

Drought, issues with crop quality and production, and challenges with yield forecast [1] are just some of the issues that modern agriculture must contend with. According to United Nations' Food and Agriculture Organization (FAO) estimates, the global population will reach 8 billion by 2025, an increase of three people per second (or 250,000 people per day), and rise to an estimated 9.6 billion by 2050 [2]. Farmers need to drastically expand food production while protecting the environment and efficiently using natural resources to keep up with the growing global population. Still, they can not achieve this independently, and traditional farming methods are not equipped to handle the task. To that end, technological advancements are crucial in meeting the global population's expanding dietary requirements.

Farmers need to know how much water their crops will use, when that water will be used, and how much water will evaporate from the surface of lakes and dams for free [3]. The ultimate goal of smart irrigation management is to achieve efficient and effective water use, making it an increasingly significant administrative activity. Good irrigation management aims to apply the appropriate amount of water at the appropriate time.

New and improved technologies are helping the vast majority of people in many different ways. The Internet of Things (IoT) and data analysis through machine learning [4] significantly enhance people's ability to shape their environments in their daily lives. IoT and machine learning are widely used in the agro-industrial sector for diagnosing and managing smart irrigation systems, both of which supply crucial data for successful farming.

Smart irrigation that efficiently uses water could benefit from IoT-based solutions, which have already proven useful in many areas of the agricultural environment [5]. An intelligent irrigation system must consider soil moisture, precipitation, and evaporation.

A smart irrigation system is proposed in this paper, which would utilize machine learning algorithms and the IoT to conserve water and power. Soil moisture, temperature, humidity, and rainfall are just some variables the system's array of sensors may track. A machine learning system is fed data from these sensors about the crop moisture level to formulate an ideal irrigation schedule. In addition, the system incorporates IoT devices, which provide full transparency by enabling remote user control and system monitoring.

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The smart irrigation system is effective because it uses machine learning algorithms to analyze crop data in real time and predict plant development and harvest. The benefits of the smart irrigation system are discussed in this study as well, including decreased water consumption, increased crop production, decreased energy consumption, and decreased operational expenses.

The rest of this paper is organized in the following manner. Section 2 reviews the related work and section 3 presents the proposed methodology in detail. While section 4 presents the experimental evaluation, section 5 presents the system prototype. Finally, the paper is concluded in section 6.

II. Related Work

IoT, machine learning capabilities, data analytics and mining [6], and wireless technologies [7] contribute to developing smart agriculture management. Many works presented a lot of suggestions as well as several approaches to general agricultural activities that make use of this new technology.

In [8], an overview of agricultural irrigation systems that use the Internet of Things is presented. In their literature study, the authors analyze wireless technologies, sensor nodes, and IoT systems alongside the most important indicators for characterizing water quality in irrigation, soil, and weather situations [9] and discuss the methods used to monitor and manage automated watering systems like sprinklers. In order to attain precision irrigation for increased food safety and water conservation, the authors offer new insights into how existing monitoring and advanced control systems might be developed. Critical agricultural applications are identified and examined, and the potential of IoT technology, practices, and future research are all explored in [10]. The research projects reviewed the state of the art in smart agriculture monitoring in both laboratory and field settings.

In [11], a mathematical model was provided for corn tempering drying that considers the thermo-physical heterogeneity of the kernel's components. In [12], an infrastructure for implementing smart irrigation systems was presented, utilizing LoRaWAN and fog computing. In order to determine when to irrigate, the proposed system exploited IoT nodes that communicate with nearby fog computing nodes and a distant cloud. No system has been proposed where data can be stored and tracked locally. According to the results of their analysis, cloud computing is more expensive.

In [13], a Windows GUI program was described for multifractal analysis of soil and plant structures. In [14], a user-friendly smartphone app, REUTIVAR-App, was introduced for fertigation scheduling in olive trees with recycled water. Farm-scale irrigation and fertilization recommendations are also tracked daily in real-time as part of their research. However, the study did not focus on IoT uses.

In proposing an IoT platform, the authors in [15] showed how to collect, analyze, and visualize sensor data for use in smart farming. In [16], a proposal was made for monitoring environmental parameters in chicken farms using inexpensive gear and software. Several experiments were also conducted to contrast the proposed prototype with existing commercial options. Intending to realize long-term and cost-effective monitoring, the authors in [17] proposed a soil environment monitoring system based on RFID sensors and LoRa for tracking soil's temperature, moisture level, and chloride ion concentration.

The authors of [18] present a method for recommending when and how much water should be used for irrigation by combining agglomerative clustering with machine learning methods like regression trees and support vector machines. The recommendation system did well on its collected data set and the publically available agricultural data set. The authors of [19] created an IoT and a data-driven irrigation system. Predictions of soil moisture content are performed using this method and compared to those made using alternative state-space models. Results show that the suggested strategy outperforms the status quo, and predictions for soil moisture content are positive.

Since a water resource is most misused in irrigation, monitoring and controlling the agricultural field or irrigation are key factors to be met in an efficient, smart irrigation system [20]. The widespread availability of mobile phones makes this possibility possible. For off-site, real-time field data monitoring, an IoT-based strategy was developed and operated [21]. In [22], the authors proposed a system that used the ThingSpeak cloud platform based on the IoT to enable remote monitoring and control of the irrigation system. By allowing for remote monitoring and control, the system made the farmer's job easier in terms of optimizing the irrigation system. Incorporating foresight utilizing machine learning and artificial intelligence is a hot topic of study. Modern irrigation techniques are necessary to prevent water shortages [23]. The next section details the suggested methodology and the basic prerequisites needed to implement it.

In order to predict the agronomist's irrigation suggestions, this study [24] set out to establish a model that incorporates his irrigation planning process. Machine learning techniques applied to an integrated dataset from multiple sources allowed us to accomplish our goal. The dataset included agronomist-determined irrigation levels and sensor-based characteristics, weather-based features, and features describing plots. Three machine learning techniques, the standard linear regression, the nonparametric gradient boosted regression trees, and the nonparametric boosted trees classifiers, were used to build various irrigation recommendation prediction models utilizing this comprehensive dataset. Eight distinct feature subsets were used to train the models.

Predicting agricultural yields and water availability using a hybrid of the wrapper feature selection methodology and the PART classification technique is offered [25], called WPART. There are four phases to the proposed system: the first is data collection, wherein environmental indicator data is gathered via the IoT system; the second is data preprocessing; the third is feature selection, wherein the wrapper feature selection approach is used to analyze environmental indicators and select the effective indicators of the farming system; and the fourth is drought prediction, wherein PART supervised machine learning algorithms are used to analyze the selected indicators.

III. The Proposed System Architecture

Figure 1 depicts the proposed system's design. There are three phases: Data collection, ML model building, API integration, and irrigation scheduling. The watering scheduler is improved by the data fusion model used in the cloud service. The data model can be customized for various use cases by selecting and feeding in data from specialized domains of knowledge. In addition, the watering schedule may be monitored and adjusted remotely using a web interface.

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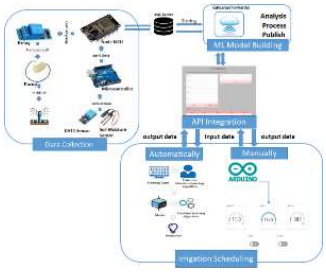


Fig. 1. The proposed system architecture

A. Data collection

The smart irrigation system comprises various components for efficient data collection. Sensors such as a soil moisture sensor and a DH11 sensor for temperature and humidity air measurements are deployed in the field. These sensors are connected to an Arduino Uno microcontroller, which serves as the central control unit. The microcontroller collects real-time data from the sensors, facilitating data processing and communication with other system components. A Node MCU is also utilized to establish an internet connection, enabling data transmission to the cloud. The collected sensor data, including soil moisture levels, temperature, and humidity, is stored in a database server, providing a reliable and scalable solution for data management.

B. ML model building

Once the data is collected from the sensors and stored in the database server, it is transmitted to the cloud for processing and analysis. In the cloud-based environment, data undergoes rigorous preprocessing to ensure its optimal quality and relevance, a crucial step given the significance of smart irrigation systems. The initial phase involves addressing missing values, often a result of intermittent connectivity or sensor malfunctions. To achieve uniformity in the disparate measurements, data such as temperature and humidity are standardized using Min-Max scaling. Further, categorical variables, like soil types, undergo a transformation process called one-hot encoding, making them digestible for machines without implying any ordinal significance. A pivotal step in the preprocessing journey is the partitioning of data into training and test subsets. This division is paramount to avert overfitting, a situation where a model might excel on its training data but falter drastically with unfamiliar data. By training the model on one subset and testing it on an unseen one, a more realistic gauge of the model's performance in real-world scenarios is achieved. Once preprocessed, the data is utilized to train a machine learning model hosted in the cloud, with the test data serving the purpose of model evaluation. The cloud platform also hosts an API for users to access system functionalities. It stores and manages the system's data, ensuring scalability, reliability, and security. Additionally, it enables integration with other services and applications, acting as a centralized hub for data processing, analysis, and decision-making in the smart irrigation system. Once the collected sensor data goes through a comprehensive analysis process in the cloud platform, the smart irrigation system leverages the power of machine learning to ensure accurate and efficient irrigation decisions. A labeled dataset is then prepared, incorporating historical sensor data and corresponding irrigation procedures, providing valuable insights into when to activate or deactivate irrigation based on sensor readings. The machine learning model is trained to capture the intricate relationships between sensor readings and irrigation requirements using sophisticated machine learning algorithms such as decision trees (DT), support vector machines (SVM), or logistic regression (LR). Once trained, the model is deployed in the cloud platform, enabling real-time inference for predicting optimal irrigation actions based on the learned patterns and relationships. This intelligent decision-making significantly improves water efficiency and empowers farmers with precise and timely irrigation strategies.

C. API integration

API integration is critical in enabling real-time decision-making for irrigation control in a smart irrigation system. By integrating machine learning models with a cloud server through APIs, the system can harness the power of data analysis and predictive algorithms to optimize irrigation operations. Users can retrieve sensor readings, watering recommendations, and system configurations through an API. It allows users to access real-time data, monitor sensor readings, configure irrigation settings, and switch between manual and automatic modes. The API provides a unified interface, enabling seamless integration with external services or applications. This integration allows users to interact with the system and make informed decisions. The API serves as a key component to share data, control the system, and facilitate smart irrigation system integration with other systems or applications.

D. Irrigation scheduling

The smart irrigation system works automatically by harnessing the power of machine learning. By integrating machine learning models with the system, it is automatically decides to pump or not without user intervention.

In addition to its automatic capabilities, the smart irrigation system also provides manual control through an IoT cloud Arduino service, such as a web application. Users can access the web application interface, which allows them to monitor sensor readings, adjust irrigation settings, and manually control irrigation actions. By selecting the manual mode and using the web application, users can override the automatic irrigation decisions made by the machine learning models. They have the flexibility to customize irrigation parameters based on their preferences or specific crop needs. For instance, users can choose to activate or deactivate the irrigation system, adjust the watering duration, or set specific irrigation schedules. The manual control feature provides users with a higher level of control and adaptability, allowing them to fine-tune the irrigation process to suit their specific requirements or respond to unexpected environmental changes.

IV. Results and Discussions

A. Dataset profiling

The dataset consists of six attributes; CropType, CropDays, SoilMoisture, Temperature, Humidity, and Irrigation. The dataset attributes description is depicted in Table 1. The dataset "Crop Irrigation Scheduling," used in this work, was downloaded from Kaggle [26].

Table 1. The dataset attributes description

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Attribute	Description
CropType	This attribute indicates the specific type or species of crop being cultivated in the agricultural field. Purpose: Knowing the crop type is crucial because different crops have varying water and nutrient requirements. It helps tailor irrigation strategies and optimize crop yield.
CropDays	CropDays represents the number of days or a time-related metric since the planting or germination of the crop. Purpose: CropDays is essential for tracking the growth stage of the crop. It helps in determining when specific growth-related activities or irrigation interventions might be required.
SoilMoisture	Soil moisture is a measurement of the moisture content in the soil, often expressed as a percentage. Purpose: Monitoring soil moisture levels is critical for understanding the current state of soil hydration. It guides irrigation decisions by indicating whether the soil needs water or not.
Temperature	Temperature refers to the ambient air temperature in the vicinity of the agricultural area, often measured in degrees Celsius or Fahrenheit. Purpose: Temperature affects plant growth and evaporation rates. It helps in assessing the climate conditions and determining the suitability for irrigation.
Humidity	Humidity represents the amount of water vapor in the air, typically expressed as a percentage. Purpose: Humidity levels impact the rate of moisture loss from both the soil and the crop. High humidity can reduce evaporation, affecting the timing of irrigation.
Irrigation	The Irrigation attribute typically indicates whether irrigation was performed, and it might include additional information about the amount or duration of irrigation. Purpose: Irrigati on data tracks the actual application of water to the crop. It is essential for assessing the effectiveness of irrigation practices and ensuring adequate water supply to the crops. These attributes collectively form the dataset used in a smart irrigation system. By analyzing and making decisions based on this data, the system can automate irrigation processes, conserve water resources, and optimize crop growth, leading to improved agricultural outcomes.

Irrigation decisions are crucial in agriculture to ensure that crops receive the right amount of water at the right time. This attribute may contain data or actions related to irrigation management.

These attributes collectively provide essential information for managing and optimizing crop cultivation, irrigation, and overall agricultural practices. They are valuable for making data-driven decisions in agriculture and optimizing crop yield and quality.

B. Results

When comparing the results of decision trees, logistic regression, and vector machine support models, their performance based on accuracy, recall, F1 score, and accuracy are analyzed. The experimental evaluation method is the confusion matrix shown in Fig. 2 for the classifier decision tree, as an example. The mentioned evaluation measures are computed from the confusion matrix [27].

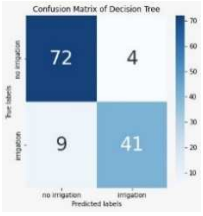


Fig. 2. The confusion matrix of the decision tree classifier

The values of computed evaluation measures, accuracy, precision, recall, and f1 score are depicted in Table 2.

TABLE 2. THE PERFORMANCE MEASURES OF DIFFERENT CLASSIFICATION MODEL

ML Model	Accuracy	Precision	Recall	F1 score
LR	0.8514	0.9189	0.7391	0.8192
DT	0.8968	0.9111	0.82	0.8631
SVM	0.8415	0.8571	0.7317	0.7894

Based on these comparisons, the DT model performs better overall, achieving high precision, recall, and F1-score for the results as shown in the Fig. 3.

In the development of our smart irrigation system using IoT and machine learning techniques, the Decision Tree classifier played a pivotal role in decision-making processes, achieving the highest performance. While various training parameters can be adjusted to refine the performance of Decision Trees, for the scope of this study, we relied on the default parameters to ensure consistency and replicability. Specifically, the criterion for the quality of a split was set to "gini" which measures the impurity of the data, the maximum depth of the tree was left undefined allowing the tree to expand until all leaves are pure, the minimum samples split was set to 2, and the minimum samples leaf was set to 1. By employing these default values, a foundational benchmark for the system's performance is established, which can serve as a reference point for future iterations and optimizations.

C. Comparison with literature

This section presents and discusses a comparison with work in the literature that is related to our proposed method. In the realm of agri-tech, our work underscores the promise of combining IoT with machine learning. Iorliam et al. [28] tested various models on IoT-based irrigation data, where logistic regression reached 71.76% accuracy, and the support vector machine achieved 90.21%. Interestingly, our decision trees model's 89.68% accuracy is nearly on par with their SVM results. Tas et al. [29] crafted a versatile, efficient irrigation system. Using sensors to track soil humidity, temperature, and rain, they tested several models. The KNN model was most impressive, with a 98.3% accuracy and 0.12 RMSE. Notably, their dataset differs from ours. Research [30] proposed an advanced Irrigation Water Quality Prediction (IWQP) model utilizing convolutional neural architectures. When stacked against Logistic Regression (LR), Support Vector Regression (SVR), and k-Nearest Neighbor (kNN), their approach leaned more towards traditional machine learning, aligning with low-capability IoT devices. Mishra and Somkunwar [31] advocated for logistic regression in smart irrigation, considering various parameters. In comparison, our various models, including logistic regression, could be marginally (0.0553%) more precise.

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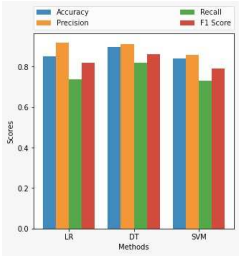


Fig. 3. The comparison between DT, SVM and LR

V. System Prototyping

As shown in Fig. 4, a system prototype is presented that synergizes IoT technology and machine learning to optimize water usage and augment crop yield. By seamlessly integrating sensors, a microcontroller, and cloud connectivity, the prototype continuously monitors and analyzes environmental data to generate adaptive irrigation schedules. Through real-time decision-making and remote control functionality, the objective is to enhance sustainability and efficiency in agricultural practices. The prototype of the smart irrigation system was meticulously implemented in a small house garden, effectively demonstrating its practicality and efficacy within a confined space. The primary aim was to develop a compact and efficient system that could be effortlessly installed and operated by homeowners. The smart irrigation system comprised the following components:

First: Hardware Components

Microcontroller: A microcontroller, such as Arduino, was selected as the central processing unit of the system. It established seamless communication with the sensors, executed the irrigation control algorithm, and orchestrated the overall functionality of the system.

IoT Connectivity: The prototype employed a wireless module, specifically NodeMCU ESP8266, to establish robust connectivity between the microcontroller and the Internet/cloud platform. This wireless integration facilitated uninterrupted data transmission and remote access to the system.

IoT Sensors:

- Soil Moisture Sensor: Strategically positioned soil moisture sensors accurately measure the moisture content in different areas of the garden. Real-time data from these sensors served as vital input for the irrigation control algorithm.
- Temperature and Humidity Sensors: Additional sensors were deployed to meticulously monitor environmental conditions, including temperature and humidity. These sensors provided essential information for determining optimal watering requirements based on prevailing weather conditions.
- Water Pump: A compact water pump was seamlessly integrated into the system to ensure efficient water delivery to the plants. Upon detecting the need for irrigation based on sensor readings, the microcontroller promptly activated the water pump to supply water to the garden.
- Power Supply: A consistent power supply was ensured to facilitate uninterrupted operation of the microcontroller and sensors.

Second: Software Components

- Machine Learning Algorithms: The prototype harnessed advanced machine learning algorithms to effectively analyze sensor data and make accurate predictions and decisions regarding irrigation schedules. These algorithms were meticulously trained using historical data to discern irrigation patterns and optimize watering schedules based on the unique requirements of different crop types.
- Arduino IoT Cloud: The Arduino IoT Cloud platform served as the central hub for efficient data management and seamless system control. It empowered users with real-time monitoring capabilities, robust data storage, and remote access to system configurations through an intuitive user interface.
- API Integration: To enhance system flexibility and enable seamless interaction with external systems, we developed a custom API. This API facilitated effortless customization, scalability, and potential integration with other agricultural management systems. Its functionality included facilitating the retrieval of sensor readings, configuration of irrigation settings, and real-time interaction with the system.

During the development of the prototype, we meticulously integrated the hardware and software components to create a cohesive and seamlessly functioning system. The microcontroller served as the intelligent core, effectively receiving data from the sensors and executing the irrigation control algorithm based on the trained machine learning models. IoT connectivity enabled robust communication between the microcontroller and the cloud platform, facilitating uninterrupted data transmission and remote access to the system. The software components, comprising machine learning algorithms and the Arduino IoT Cloud, worked synergistically to analyze sensor data, make real-time decisions, and provide users with a user-friendly interface for monitoring and controlling the irrigation system.

The seamless integration of the hardware and software components ensured the smooth operation of the prototype. Sensor data collected by the hardware components served as critical input for the machine learning algorithms, which generated precise irrigation schedules tailored to the specific requirements of different crop types. The Arduino IoT Cloud platform facilitated real-time monitoring, data storage, and remote access, empowering users to interact with the system and make necessary adjustments as per their needs. The API integration further enhanced system flexibility and potential for integration with other agricultural management systems.



Fig. 4. The prototype of the proposed system

VI. Conclusion

In conclusion, this research highlights the significance of incorporating technological advancements, such as IoT and machine learning, in addressing the challenges faced by modern agriculture. The proposed smart irrigation system offers an innovative solution to optimize water usage, increase crop production, and reduce operational expenses. By leveraging sensor data, cloud IoT platforms, and machine learning algorithms, the presented system provides real-time

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and accurate irrigation recommendations based on the specific needs of crops. It allows for effective water management, considering factors like soil moisture, temperature, and humidity. The integration of machine learning models enhances the system's decision-making capabilities, enabling adaptive and precise irrigation scheduling. The decision tree algorithm has the highest accuracy (0.8968), followed by the logistic regression algorithm (0.8514) and the support vector machine (0.8415) in this comparison of machine learning techniques' efficacy. In comparison to logistic regression and support vector machine techniques, the decision tree approach has a greater precision (0.0553 vs. 0.0454). The prototype implementation in a small garden further validates the practicality and effectiveness of the system in promoting water conservation and supporting healthy plant growth.

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